
Professional Summary

Mechanical Engineering student (4.0 GPA) with an entrepreneurial drive, specializing in mechatronics, First Principles design, CAD/FEA, and Design for Manufacturability (DFM). Combines a rigorous academic foundation in statics, dynamics, and structural mechanics with extreme ownership and hands-on operational experience. Adept at designing autonomous system housings, optimizing actuator/motor integration, and solving complex center-of-gravity challenges. Highly motivated to apply structural optimization and robotics engineering expertise to accelerate automation.

Qualifications & Technical Skills

Engineering SolidWorks (Simulation/FEA), First Principles Analysis, AutoCAD, MATLAB, Arduino IDE.

Robotics Kinetmatics & Dynamics, Actuator/Motor integration, Sensor Packaging, Center of Gravity

Manufacturing Design for Manufacturability (DFM), GD&T, Rapid Prototyping, BOM Management.

Ops/Spatial Dynamic Load Management, Real-world Weight Distribution, High-Stress Physical Operations.

Tech/Software Java, C/C++ (Arduino), Web Development (HTML/CSS/JS), Data Analytics.

Engineering, Operations, & Entrepreneurial Experience

Fall 2025 – Present **Mechanical Designer (Autonomous Robotics & Systems), UIC, Chicago, IL**

- Engineered custom mechanical assemblies utilizing SolidWorks and First Principles analysis, optimizing the structural packaging of brushless DC motors, batteries, and sensitive autonomous computing hardware.
- Executed rigorous FEA and mass-reduction strategies on mechanical frames to ensure optimal dynamic stability and structural integrity during autonomous operational cycles.
- Spearheaded cross-functional integration with electrical teams, architecting a scalable Bill of Materials (BOM) to streamline the manufacturing of precision mechanical housings tailored for complex sensor arrays.

Jan 2026 – Present **Founder & Digital Consultant, Freelance Web & Ad Agency, Chicago, IL**

- Demonstrate extreme ownership by independently architecting, developing, and deploying high-performance digital infrastructure for local businesses to drive customer acquisition.
- Engineer and manage targeted digital advertising campaigns, utilizing data analytics and A/B testing to optimize performance metrics—mirroring the rapid, data-driven iteration required in complex systems design.
- Interface directly with stakeholders to translate complex requirements into deployed technical solutions.

April 2026 –
Present

Cargo Agent (American Airlines Contractor), Unifi / American Airlines, Chicago, IL

- Manage the physical staging and loading of 20+ tons of commercial freight daily, enforcing strict center-of-gravity, weight distribution, and dynamic load limits crucial for maintaining spatial stability.
- Navigate high-stress operational environments, maintaining extreme spatial awareness and precision logistics analogous to the physical constraints and balancing requirements of complex bipedal/mechanical systems.
- Collaborate with cross-functional ground crews to safely turn 10+ heavy-logistics operations per shift, analyzing physical bottlenecks to improve turnaround efficiency.

Oct 2025 – Present **AI Trainer & Prompt Engineer (STEM Focus), Handshake & LinkedIn AI Project, Remote**

- Evaluated and optimized Large Language Model (LLM) outputs in specialized STEM domains, ensuring high-fidelity reasoning in physics, kinematics, and structural mechanics applicable to autonomous problem-solving.

Education

Expected May 2028 Bachelor of Science in Mechanical Engineering, University of Illinois at Chicago (UIC), Chicago, IL, GPA: 4.0/4.0
Relevant Coursework: Thermodynamics, Strength & Materials, Statics, Dynamics, Structural Analysis, Engineering CAD Design.

Affiliations

Present American Society of Mechanical Engineers (ASME) – Member